Beyond concordance: External validation of probabilistic grammar

in the English dative alternation

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Abstract

This paper treats the external validation of a corpus-based probabilistic grammar as a full predictive-evaluation problem, with the English dative alternation as a worked case. Earlier studies already train a dative model on one variety and apply it to another, reporting the transfer as discrimination or classification accuracy. I freeze a production model fit to the `languageR` dative data of Bresnan et al. (2007) and evaluate its move to the later Spoken British National Corpus 2014 (BNC2014) conversational data with proper scoring, calibration, a paired target-native benchmark scored on the same observations, predictor-group ablation, missing-data sensitivity, and an explicit opportunity-set denominator. The source-trained profile retains substantial discrimination in the new domain, and the portable part is mainly non-lexical conditioning rather than lexical base rates; but calibration and the paired native benchmark expose domain-specific structure that discrimination alone hides. The fitted quantity is a conditional production probability inside an annotated alternation sample, not a probability of grammaticality, so the validated model projects within a stated target domain while staying silent, by construction, about the grammatical-but-unattested region. The result is a validation ladder for probabilistic grammar and an account of what each rung licenses.

Keywords: dative alternation; probabilistic grammar; external validation; cross-corpus transport; calibration; projectibility; corpus linguistics

1 The problem

A corpus-based probabilistic grammar can predict held-out observations from its own source corpus and still fail when it is moved to another corpus. External validation therefore needs more than a single discrimination statistic. It needs probability scoring, calibration, comparison with target-native predictions on the same observations, an account of which predictors carry the transport, and a specification of the opportunity set over which the predictions are defined. This paper assembles

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that regime and runs it on a canonical case. Throughout, "probabilistic grammar" names the fitted production model, not a claim about speakers' competence or a community's licensing of forms; external validation bears on the model's predictive transport, not directly on either.

The English dative alternation is that case. Speakers can say "give Kim the book" or "give the book to Kim", and the choice is uneven: it varies with verb, length, animacy, definiteness, pronominality, discourse status, corpus, and register. The unevenness is easy to misread, because a rare pattern can be lexically restricted, strongly dispreferred, or tied to a discourse context without falling outside the grammar. Bresnan et al. (2007) modelled that conditioning and argued that corpus probabilities can reveal grammatical possibilities informal judgements miss.

The model has since been shown to travel: trained on one variety and applied to another, it keeps much of its accuracy (Bresnan & Hay, 2008; Engel et al., 2025; Kendall et al., 2011). The open question is not whether it travels but how well it travels once transport is treated as a full predictive-evaluation problem, and what a fitted production probability can and cannot say about grammar. That is a question of PROJECTIBILITY: what observing a conditioning profile in one setting licenses us to predict about another (Goodman, 1955). The paper's discipline is to keep the settings, and the quantities they estimate, apart.

2 The original target

Bresnan et al. (2007) is the anchor because it treats the alternation as a grammatical problem rather than as a simple frequency count. The claim is not that whatever occurs often is grammatical. The claim is that observed choices can expose systematic constraints on what speakers treat as available.

That move matters because informal judgements often compress several facts into one verdict. A construction may sound odd because it's impossible, because it is rare, because the wrong verb is present, because a discourse condition is missing, or because the constructed example strips away the context that would license it. Corpus data can help separate these cases.

The empirical target is production choice. In Bresnan et al.'s setup, the response variable is whether the recipient appears as a noun phrase (NP) in the double-object pattern or as a prepositional phrase (PP) in the prepositional dative. The predictors describe properties of the verb, recipient, theme, discourse status, and corpus source.

This paper preserves that target. It treats production probability as evidence to be interpreted, rather than as a direct measure of grammaticality. The empirical question is whether the production profile survives cleaner modelling and whether it travels beyond the original corpus setting.

3 Data reproduction and reanalysis

The analysis starts with `languageR::dative`, distributed with the Comprehensive R Archive Network (CRAN) `languageR` package (Baayen & Shafaei-Bajestan, 2025). The source tarball supplies the packaged `dative.rda` object; the analysis scripts load it and write derived summaries.

The package documentation identifies the sources as Switchboard and the Treebank Wall Street Journal collection. Those sources place the original data in American English: Switchboard is telephone conversation among speakers from the United States, developed in 1990–1991 (Godfrey & Holliman, 1993), and the Treebank Wall Street Journal component is late-1980s American news and financial writing, with one million words of 1989 material and a larger 1987–1989 Wall Street Journal source collection (Charniak et al., 2000; Marcus et al., 1999).

In the packaged dataset, there are 3,263 rows and 15 columns. The response records recipient realization, with 2,414 NP realizations and 849 PP realizations. The dataset contains 75 verb levels and two modalities: 2,360 spoken tokens and 903 written tokens. Speaker identifiers are missing exactly for the written rows.

For the first model sequence, the response stays binary: NP is coded as 1 and PP as 0. The initial split is verb-stratified, with a fixed seed (`20260623`), so every verb observed in the test set is also represented in training. That split gives 2,637 training rows and 626 test rows.

The denominator is therefore narrow. These rows are already tokens in which a dative recipient was realized as NP or PP. They aren't all contexts in which a transfer event could have been expressed, omitted, paraphrased, or packaged in some other construction. The models estimate conditional choice within an attested alternation frame: given that a token enters the dative-alternation sample, which realization appears?

Initial analyses are deliberately plain. A marginal model sets the baseline. Structured models then add modality, length, verb, and the non-verb predictors. Negative-control and simulation checks accompany them: a scrambled-label check and fake-data checks. These tests ask whether the fitted structure beats chance, not whether it proves a grammatical theory.

A second empirical source is the Spoken British National Corpus 2014 (BNC2014) dative dataset (G. Jenset & McGillivray, 2018; G. B. Jenset & McGillivray, 2019). The verified Figshare comma-separated values (CSV) file has 1,839 parsed data rows and 44 columns under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence. The apparent line-count discrepancy is mechanical: the file has 1,839 newline characters but no terminal newline, giving 1,840 logical text lines including the header.

The BNC2014 dative file comes from the Early-Access Subset of the Spoken BNC2014, whose spontaneous conversations were recorded in 2012–2014 and contain speaker and context metadata.

BNC2014 serves as a transport target rather than a second copy of `languageR::dative`. Its outcome field is `Pattern`, where the verb–noun–noun code `VNN` maps to NP realization and the verb–noun–prepositional-phrase code `VNPP` maps to PP realization. The six shared verbs are *give*, *send*, *show*, *sell*, *offer*, and *lend*. All six occur in the packaged `languageR` dataset.

4 Modern modelling update

The `languageR` result is a robustness result. The central question is whether modern modelling changes Bresnan et al.'s linguistic conclusion or mainly repackages it statistically. The answer, so far, is the latter. The table reports log loss, area under the receiver operating characteristic curve (AUC), and a compact parameter count.

The non-verb `languageR` formula uses modality, semantic class, recipient length, recipient animacy, recipient definiteness, recipient pronominality, theme length, theme animacy, theme definiteness, theme pronominality, and recipient/theme accessibility. The fixed-verb model adds verb as a fixed effect. The hierarchical model uses the same non-verb fixed effects and a varying verb intercept. All are logistic models for NP versus PP realization with R treatment contrasts and no interactions.

The non-verb model already predicts held-out production choices well. Adding fixed verb effects improves test log loss from 0.271 to 0.234 and raises AUC from 0.932 to 0.951. The improvement confirms that lexical conditioning is not a minor residue after length, animacy, definiteness, pronominality, and accessibility are included.

Model	Test log loss	Test AUC	Param.
Marginal null	0.554	0.500	1
Non-verb main model	0.271	0.932	18
Fixed-verb full model	0.234	0.951	92
Hierarchical verb-intercept model	0.233	0.952	19

Table 1: Held-out `languageR::dative` results for the main model sequence.

The hierarchical verb-intercept model reaches almost the same held-out performance with a smaller fixed-effect parameterization. Its test log loss is 0.233 and its AUC is 0.952. The table's parameter count for this model is the 18 fixed-effect coefficients plus the verb-intercept variance parameter; it is not an effective degrees-of-freedom estimate for the 75 partially pooled verb effects. The estimated verb random-intercept standard deviation is 2.106, so the lexical signal remains large under partial pooling.

The negative-control and simulation checks behave as they should. Scrambled-label fits lose predictive value. Fake-data fits without a verb signal collapse toward singular zero-variance random-effects models, while a fake verb-effect check recovers a nonzero verb standard deviation. These checks make the production result more credible, but they do not change its explanatory level.

The result supports Bresnan et al.'s production-choice conclusion. It does not create a new grammatical conclusion on its own. The same-data analysis shows that the classic profile is reproducible and statistically stable. The paper's new empirical pressure has to come from transport.

5 Cross-corpus transport

The BNC2014 analysis asks whether the production profile travels across corpus design, variety, period, and annotation regime. The transport model is trained on the spoken `languageR` rows restricted to the six BNC2014 verbs. It is then scored on harmonized BNC2014 rows, with complete-case filtering stated as part of the estimand.

Two corpus targets are kept apart. The `languageR` models estimate conditional production choice among attested NP and PP dative tokens. The BNC2014 analysis estimates out-of-domain predictive performance over complete-case rows in the six shared high-frequency verbs after harmonization. These are different evidence channels, not repeated measurements of one quantity. A third channel, controlled acceptability preference, is a different target again and is kept to the supplementary DAIS bridge.

Out-of-domain transport of a dative model is not new ground either. Bresnan and Hay (2008) extended the Bresnan et al. (2007) analysis to spoken New Zealand English; Kendall et al. (2011) trained a model on the Bresnan et al. (2007) data and predicted the alternation in African American English, then reversed the direction, reporting concordance indices of 0.97 and 0.95; and Engel et al. (2025) trained separate English and Dutch models and tested each on the other language. Bresnan and Ford (2010) and Röthlisberger et al. (2017) document broad cross-variety stability by other means. So neither the finding that dative conditioning travels nor the act of scoring one variety's model on another is the contribution here.

What those transport studies report is discrimination: a concordance index, equivalently the area under the curve, or classification accuracy, that is, whether the moved model ranks or labels new tokens correctly. They do not ask whether it stays calibrated, how it compares to a target-native model fit and scored on the same observations, which predictor groups carry the transport, how missing data and the harmonized denominator bound the claim, or what quantity the fitted probability estimates. This paper treats external validation as that fuller problem. The contribution is the validation regime: proper scoring, calibration intercept and slope, a paired target-native benchmark on identical rows, predictor-group ablation, missing-data sensitivity, and an explicit opportunity-set denominator, with an open harmonization record that lets each step be checked.

The comparison is a bundled out-of-domain test. The source side is American telephone conversation from 1990–1991, restricted to the six shared spoken `languageR` verbs. The target side is spontaneous British conversation from 2012–2014, under a different annotation scheme. Those six verbs account for 2,201 of 3,263 `languageR` rows and for 1,564 of 2,360 spoken rows. They are the high-frequency core, not a random sample of the alternation.

The transport models are logistic generalized linear models with logit link. The core formula is

$$\begin{aligned}\Pr(\mathrm{NP}) \sim\ & \text{Verb} + \text{recipient length} + \text{theme length}\\ & + \text{recipient animacy} + \text{recipient pronominality}\\ & + \text{theme animacy} + \text{theme definiteness} + \text{theme pronominality}.\end{aligned}$$

Lengths are raw word counts, with no centering, scaling, or logging. Factors use R treatment contrasts, with *give* as the verb reference and "no" as the binary reference. There are no interactions in the transport models. Factor levels and coding are fixed before scoring BNC2014; no BNC2014-specific scaling or imputation enters the headline model. Recipient definiteness is unavailable in BNC2014 and enters only as a labelled regex proxy in a sensitivity check. The validation is retrospective: freezing means the source coefficients are not re-estimated from BNC2014 outcomes, not that the target corpus was unavailable while the harmonization, common-predictor set, and six-verb restriction were chosen. Those design choices are target-aware by necessity; the fitted coefficients are not.

Harmonization is lossy. The outcome maps cleanly: `VNN` becomes NP and `VNPP` becomes PP. Verb maps cleanly after the six-verb restriction. Length needs more care. The released BNC2014 length fields are character counts: the data dictionary defines `RecLen` and `ThemeLen` as the number of characters in the recipient and theme (G. B. Jenset & McGillivray, 2019). Those fields are not comparable to `languageR`'s token-like lengths, so the scripts derive word counts from the released phrase strings. Rows with empty phrase strings cannot be used in the complete-case transport model.

The core transport model scores the 1,621 BNC2014 rows complete for the harmonized predictors, not all 1,839 released dative rows. The missing rows are outcome-skewed: NP recipients make up 0.819 of the complete rows but only 0.670 of the 218 incomplete rows. The variable-level pattern is also uneven. Recipient-pronominality is missing for 112 rows with NP rate 0.357, while theme-pronominality is missing for 139 rows with NP rate 0.950. The headline comparison is consequently a complete-case estimate over the shared high-frequency core.

A BNC2014 metadata audit makes the domain label concrete. In the released dative rows, Level 1 dialect is "uk" for 1,720 of 1,839 rows and Level 2 dialect is "english" for 1,689. Gender is female-coded for 1,138 rows and male-coded for 699, and the largest age band is 19–29 with 736 rows. Close-family, partner, and very-close-friend conversations contribute 1,171 rows. The complete-case subset has the same broad shape. A single BNC2014 transport score should therefore be read as a corpus-domain diagnostic, not as a clean estimate for British spoken English in general.

The released BNC2014 file also limits the resampling design. The supporting scripts show that an intermediate `main_speaker_id` was derived and selected, but it was dropped before the public 44-column CSV. No stable speaker, text, or conversation identifier is recoverable from the released files. The native BNC2014 benchmark therefore uses repeated verb-stratified row-level cross-validation, not grouped speaker or conversation folds. The comparison below repairs the old sample mismatch by scoring source-trained and native out-of-fold predictions on the same 1,621 rows, but it cannot remove all within-conversation dependence.

Origin	Model family	Log loss	AUC
`languageR` source	marginal	0.473	0.500
`languageR` source	verb only	0.456	0.646
`languageR` source	non-verb	0.296	0.899
`languageR` source	full core	0.308	0.906
BNC2014-native OOF	marginal	0.473	0.500
BNC2014-native OOF	verb only	0.439	0.625
BNC2014-native OOF	non-verb	0.234	0.928
BNC2014-native OOF	full core	0.227	0.939

Table 2: Same-row BNC2014 complete-case evaluation. OOF = out-of-fold. All rows are the same 1,621 BNC2014 complete-case observations; the native rows use five repeated 10-fold verb-stratified row-level cross-validation. Native AUC is averaged within each held-out fold, so an intercept-only model is 0.500 by construction; log loss is pooled. The released file carries no speaker or conversation identifier, so the folds are row-level and within-conversation dependence is not removed.

The source-trained non-lexical profile transports, but not because shared verb base rates do most of the work. The source full model improves log loss over the source marginal model by 0.165 in a paired row bootstrap ($[-0.198, -0.133]$ for full minus marginal) and raises AUC by 0.406. The source verb-only model gives only a small log loss gain over marginal (0.456 vs. 0.473), while the source non-verb model reaches 0.296 log loss and 0.899 AUC. Adding the source verb terms to the non-verb model improves ranking slightly but worsens log loss slightly (full minus non-verb $= 0.012$, bootstrap interval $[-0.006, 0.030]$). In this target corpus, the portable part of the profile is mainly the non-lexical conditioning, captured by the non-verb model, rather than lexical base-rate ordering.

The target-native model is still better. On the identical rows, the BNC2014 native full model has log loss 0.227 and a fold-averaged AUC of 0.939. Against the source full model scored on the same held-out folds, its log loss is lower by 0.081 (paired row bootstrap $[-0.104, -0.059]$) and its fold-averaged AUC is higher by 0.032 (fold bootstrap $[0.025, 0.039]$). The native non-verb model is already close to the native full model, which reinforces the same interpretation: the broad conditioning profile is portable, but the target corpus supplies local calibration and residual lexical structure.

Calibration tells the same story. The source full model underpredicts NP realization for *send*, *offer*, and *lend*, and overpredicts it for *sell*. Apparent recalibration on the same BNC2014 rows, a diagnostic upper bound rather than out-of-sample performance, gives a calibration intercept of 0.44 (standard error 0.10) under an intercept-only update and an intercept of 0.58 (0.09) with slope 0.73 (0.04) under an intercept-and-slope update, lowering log loss from 0.308 to 0.301 and to 0.290 respectively. Even that best case stays short of the target-native full model's 0.227, so the remaining gap is not removed by globally rescaling the level and slope of the source predictions; it is not only a prevalence shift.

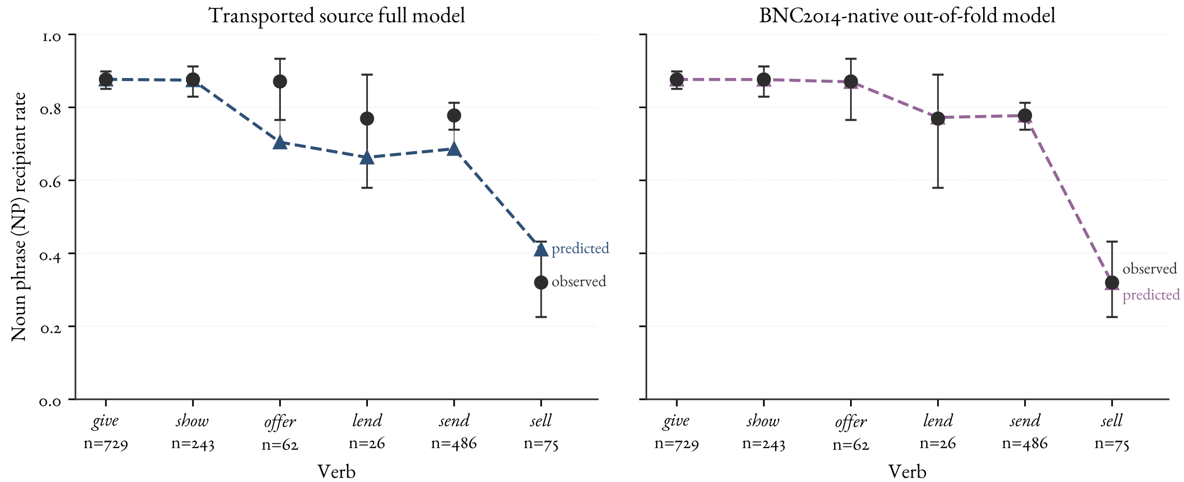

Figure 1: Verb-level calibration for the transported source full model and the BNC2014-native out-of-fold model on the same complete-case rows. Black points and intervals are observed NP recipient rates with binomial Wilson intervals, unadjusted for conversational clustering; coloured triangles are mean predictions.

A matched check separates the two things the missing rows change: the predictor set and the row denominator. Only three predictors are observed for every BNC2014 row, so the reduced common-predictor model keeps verb, theme animacy, and theme definiteness. Holding that reduced model fixed and scoring it on the 1,621 complete-case rows already collapses performance relative to the full model (log loss 0.562 vs. 0.308, AUC 0.510 vs. 0.906). Expanding the same reduced model to all 1,839 rows changes little (log loss 0.562 to 0.586, AUC 0.510 to 0.523). Dropping the recipient and length predictors, not enlarging the row denominator, accounts for almost all of the loss, which is why the headline claim stays tied to the complete-case harmonized denominator.

6 Estimand, opportunity set, and inferential scope

The transport result is informative only against a clear statement of what the model estimates. The fitted probability is, approximately,

$$P(\text{NP recipient} \mid \text{a token entered the annotated NP/PP dative sample}),$$

not the probability that a double-object form is grammatical, and not the probability that a transfer event is realized as a dative at all. The corpus rows are already tokens in which a recipient was realized as an NP or a PP. They are not the set of contexts in which a transfer event could have been expressed, paraphrased, left implicit, or judged. The models estimate conditional choice inside an attested alternation frame.

This bounds what successful transport licenses, and it separates two notions that sit close. The conditioning relations the profile encodes, the effects of pronominality, length, animacy, and definiteness, are projectible in Goodman's sense (Goodman, 1955): the kind of generalization that

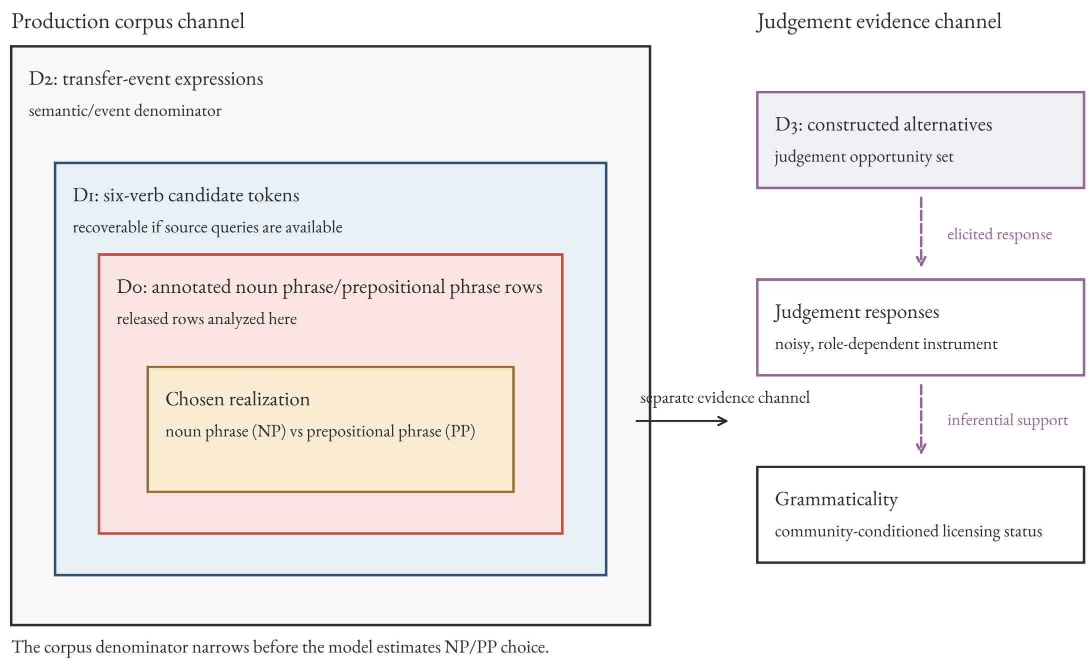

Figure 2: Nested denominators for the dative estimand. The released corpus rows fix realization choice inside the annotated noun phrase/prepositional phrase sample, the innermost set. The broader transfer-event and judgement sets are different denominators that answer different questions; reaching them is an inferential step to a new target, not a larger sample of the same one.

supports projection to new cases. Whether a fitted model then transports to a given new corpus is a separate, empirical question, and a projectible relation can still fail to transport under a population shift. Within its demonstrated scope the production model forecasts realization for unseen tokens among the verbs, registers, and varieties it covers, and it stays silent, by construction, about the grammatical-but-unattested region. No amount of additional positive-token data collected under the same sampling design D_0 can identify that region by itself; more tokens enlarge the attested NP/PP area without changing the opportunity set. That is an observational-identification claim, not a verdict that production evidence is useless. A broader denominator, elicited production, novel-item productivity tests, or acceptability judgements can each add information about the otherwise empty cells.

The acceptability channel illustrates the last route and its limits. A model validated for production choice is not thereby validated for judgement: the supplementary DAIS bridge shows the two channels separating, most clearly for *offer*, where participants accept the double-object form far more readily than the production probabilities imply. Validity is relative to a target; carrying it across channels is a further claim, not a free one.

External validation is itself a ladder; Table 3 lays out its rungs, and the dative shows what each reports. Source-data reproduction, within-domain validation, frozen cross-domain prediction, a target-native benchmark scored on the same observations, recalibration, and error decomposition are distinct steps, and a model can clear some while failing others. Here the source model discriminates well out of domain yet stays worse-calibrated than a target-native fit, and an intercept-and-slope recalibration does not close the gap, so the remaining gap is not removed by globally rescaling the predictions; it is not only a prevalence shift. Reporting only discrimination, as a concordance index or an accuracy, would have hidden that.

Stage	Data	Diagnostic	Licensed conclusion
Source reproduction	Source corpus	Held-out log loss and AUC against nulls	The source profile is real and reproducible
Within-source validation	Source corpus	Partial pooling, negative controls	Lexical and non-lexical signal are not overfit
Frozen external scoring	Target corpus	Log loss and AUC against the target marginal	The profile carries predictive information out of domain
External calibration	Target corpus	Calibration intercept and slope	Whether the levels are right, not just the ranking
Paired target-native benchmark	Target corpus	Same-row paired loss differences	How much local structure the source model misses
Predictor-group ablation	Both	Marginal, verb-only, non-verb, and full contrasts	Which part of the profile transports
Opportunity-set audit	Target corpus	Denominator and missingness map	What quantity the predictions estimate

Table 3: A validation ladder for a corpus-based probabilistic grammar. Each stage has its own diagnostic and licenses a different conclusion, and a model can pass some rungs while failing others.

The ladder extends to other alternations under stated preconditions: two corpora sharing a harmonizable categorical alternation, an annotated outcome, and a lexical core large enough to fit and to score on. Its limits carry over too. The comparison is testable only where the corpora intersect, so the periphery of an alternation goes unevaluated; preprocessing learned on the source must be frozen before the target is scored; and the released data set the ceiling on how honestly dependence can be handled. The Spoken BNC2014 release makes the last point concrete: it omits the speaker and conversation identifiers that grouped resampling needs, so the native benchmark can only use row-level cross-validation and cannot remove within-conversation dependence. That ceiling is a property of the data release, not a defect of the method. The row bootstrap and Wilson intervals reported above are therefore conditional row-level intervals: they reflect test-row resampling around the fitted predictions, not source-model estimation uncertainty or within-conversation dependence, which probably makes them too narrow.

So the analyses license a cautious, well-scoped claim. The Bresnan et al. production-conditioning profile is durable and partly transportable within the shared high-frequency core; its non-lexical component travels better than its lexical base rates; and a target-native model still recovers local structure the source model misses. What the validated model buys is prediction within a stated domain, plus a clear marker of the low-probability cells where grammaticality, if it is at issue, has to be settled through another channel and another paper.

7 Conclusion

External validation of a probabilistic grammar is a ladder, and the dative shows what each rung reports. The first rung holds: the `languageR` profile survives baselines, negative controls, simulation checks, and partial pooling, and the hierarchical model matches the fixed-verb model with a smaller fixed-effect parameterization. Modern modelling confirms the classic production-choice conclusion rather than revising it.

The profile then partly transports. On the BNC2014 complete-case rows, the spoken six-verb `languageR` model beats a marginal baseline when it is scored against a target-native out-of-fold benchmark on the same observations. The portable signal is mainly non-lexical rather than verb base rates: the source non-verb model nearly matches the source full model. But the target-native model is better calibrated, and an intercept-and-slope recalibration does not close the gap, which a global rescaling of the source predictions would have done were it only a prevalence shift. The verdict is bounded. Transport survives the null comparisons in the high-frequency core, while variety, period, register, annotation, and missingness stay bundled.

That contrast is visible only because the evaluation goes beyond concordance. Earlier transport studies report discrimination or accuracy; proper scoring, calibration, a native benchmark on the same rows, predictor-group ablation, and missing-data sensitivity tell a model that ranks new tokens from one that is calibrated to them. The regime, not the finding that the dative travels, is the contribution, and it extends to other alternations under the preconditions set out above.

The estimand keeps the claim honest. The fitted probability is a conditional production choice inside an annotated alternation sample, not a probability of grammaticality. A validated production model forecasts realization within a stated target domain and stays silent, by construction, about the grammatical-but-unattested region; no quantity of further positive tokens under the same design reaches it. Crossing that boundary needs a broader opportunity set or another evidence channel, such as the acceptability judgements whose production/preference divergence for *offer* is recorded in the supplementary bridge. That is a different question, and a different paper.

Data and code availability

The public working repository is `BrettRey/bresnan-dative-alternation-reanalysis`. A submission release should archive the exact commit or digital object identifier. The analysis scripts fetch public datasets to temporary locations and validate downloaded files by checksum.

File	MD5
BNC2014 CSV	1c041dbcb635855f3d8f4be9f3e1fced
DAIS item CSV	46af310f1633b8784f897e4482faed84
DAIS cleaned-judgement ZIP	e7b3ac6f2dd8e85bc93e6e20fc0f6ec1

Derived files and the source-verification note are stored in the repository; the README gives their paths. The principal reproduction commands are:

```
Rscript analysis/08_dais_acceptability_bridge.R
Rscript analysis/09_bnc2014_metadata_scope.R
Rscript analysis/10_bnc2014_paired_transport_cv.R
python3 analysis/06_denominator_and_figure_candidates.py
make
```

The cross-validation and bootstrap seed is `20260625`. Session information is written to a derived text file. Raw DAIS files aren't committed.

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Supplementary material: the DAIS acceptability bridge

This bridge is supplementary to the external-validation result. It records what happens when the validated production model meets an acceptability channel, and it is reported here for completeness rather than as part of the paper's main claim.

Acceptability evidence answers a different question from corpus production. The Dative Alternation and Information Structure (DAIS) dataset, introduced by Hawkins et al. (2020b), contains 50,000 human judgements over 5,000 dative sentence pairs and systematically varies verb, definiteness, and argument length. The repository carries a CC BY 4.0 licence, and the dataset authors have given explicit permission to reuse the judgements, so DAIS can enter as a compact bridge (Hawkins et al., 2020a).

The bridge keeps the targets separate. I score only the 150 DAIS items whose verbs overlap the six-verb production core. The same spoken `languageR` production model that was transported to BNC2014 predicts an NP recipient probability for each constructed DAIS pair; the DAIS item mean gives the corresponding human double-object (DO) preference, that is, preference for the same NP-recipient form, on a 0–1 scale. As a bridge analysis, it asks whether a production probability profile and a controlled preference profile point in the same direction.

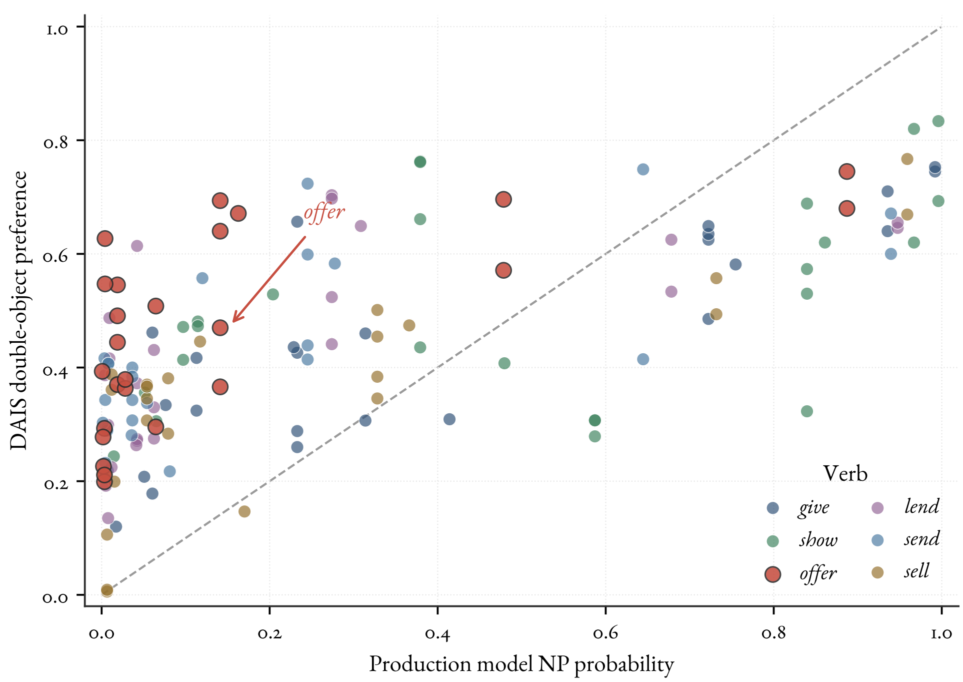

Figure 3: DAIS production/preference bridge for the six shared verbs. Each point is a constructed DAIS item. The horizontal axis is the spoken `languageR` model's NP recipient probability; the vertical axis is DAIS DO preference. The diagonal marks equality between the two channels, and the *offer* cluster is highlighted because it shows the clearest channel separation.

The two channels line up moderately, while remaining distinct. With *something* treated as a pronominal theme, Pearson's r is 0.667 and Spearman's ρ is 0.685 over the 150 shared-verb items. The

Verb	Prod. NP prob.	DAIS DO pref.	Prod. - DAIS
give	0.408	0.449	-0.041
lend	0.204	0.426	-0.222
offer	0.150	0.468	-0.319
sell	0.231	0.351	-0.120
send	0.196	0.428	-0.232
show	0.506	0.516	-0.009

Table 4: DAIS bridge for the six shared verbs. The difference column is production probability minus DAIS DO preference.

mean production NP probability is 0.283, while mean DAIS DO preference is 0.440; the mean paired difference is -0.157, with bootstrap interval $[-0.198, -0.115]$. Treating *something* as nonpronominal gives the same interpretation ($r = 0.645$, $\rho = 0.675$, mean production probability 0.343).

The item and participant checks show why the raw correlation should not be overread. At the item level, a marginal calibration regression has a positive slope, but once recipient condition, theme type, and verb are represented directly, the production-score slope is essentially zero (0.006, standard error 0.078). At the judgement level, a linear mixed model on 1,525 shared-verb judgements from 923 participants uses fixed effects for production score, recipient condition, theme type, and verb, with crossed random intercepts for participant and constructed item. The fit is non-singular, and the incremental production-score coefficient is 0.017 (standard error 0.076; approximate interval $[-0.132, 0.165]$). DAIS preference and production probability therefore show moderate marginal rank correspondence, much of it attributable to shared sensitivity to the manipulated predictors; the production score adds little once those predictors are represented directly.

The useful cases are the divergences. *Offer* is the clearest verb-level one: the production model makes it strongly PP-like, but DAIS participants are much more willing to choose the double-object version. For example, "Juan offered the woman a payment" has predicted NP probability 0.141 but DAIS DO preference 0.694. The point is channel separation: judgement preference can occupy a different region of the evidence space from corpus production probability.